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Module - 4

Introduction to ABAP in the Cloud



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Learning Objectives

- SAP BTP Overview
- ABAP Environment
- RAP (RESTful ABAP Programming)
- Extension Suite
- Extensibility
- Side-by-side Extensions

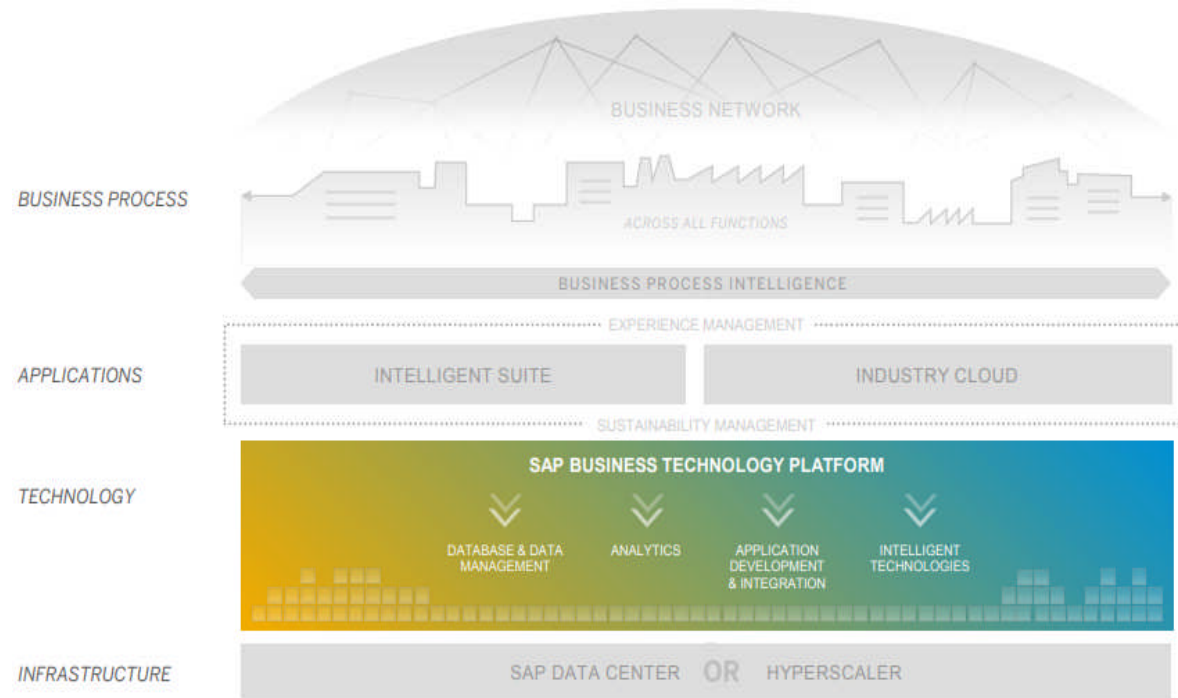


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What is SAP BTP?

- **Definition:** SAP Business Technology Platform (SAP BTP) is an integrated platform that enables businesses to turn data into business value through intelligent applications, extensions, and services. It supports innovation, integration, and data management in both cloud and hybrid environments.
- **Core Pillars:**
 - Database & Data Management
 - Application Dev & Integration
 - Analytics
 - Intelligent Technologies

SAP Business Technology Platform



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What is SAP BTP?

- **Core Pillars:**

- Database & Data Management:**

- Tools and services to manage, store, and process enterprise data — including SAP HANA Cloud, data lakes, and data virtualization.

- Application Development & Integration:**

- Frameworks and tools for building, extending, and integrating applications using various technologies like ABAP, Node.js, Java, and low-code/no-code solutions.

- Analytics:**

- Real-time analytics and data visualization using tools like SAP Analytics Cloud, enabling data-driven decision-making.

- Intelligent Technologies:**

- Incorporates AI, machine learning, IoT, and robotic process automation (RPA) to embed intelligence into business processes and applications.

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BTP Account Structure

- **Hierarchy:**

- Global Account:**

- Top-level account assigned to a customer by SAP. It represents the contract and entitlements (services, quotas, regions).

- Subaccount:**

- Logical unit within the global account used to group applications, services, and environments (like Cloud Foundry or Kyma). Can represent business units, regions, or projects.

- Spaces (in Cloud environment):**

- Subdivision within a subaccount for organizing development resources. Each space contains apps, services, and users, typically aligned with teams or stages (e.g., dev, test, prod).

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ABAP in SAP BTP

- **Definition:**

ABAP in SAP Business Technology Platform (SAP BTP) is a cloud-optimized version of the ABAP runtime environment that runs natively on the SAP HANA database. It provides the core ABAP language capabilities adapted for modern, scalable, and secure cloud development. This environment is offered as part of SAP BTP, enabling organizations to leverage their existing ABAP expertise in a cloud-native setting.

- **Purpose:**

The primary purpose of ABAP in SAP BTP is to allow developers to build cloud applications, custom extensions for SAP S/4HANA, and reusable business services. It is ideal for creating lightweight, modular solutions that integrate seamlessly with both cloud and on-premise SAP systems. By using familiar ABAP tools and development models (such as ABAP RESTful Application Programming Model – RAP), developers can rapidly deliver innovations while following SAP's clean core strategy and cloud best practices.

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Tools Used

- **Eclipse + ABAP Development Tools (ADT)**

Modern ABAP development in cloud

Integrated with SAP BTP

- **Key Features:**

Code Editor: Auto-complete, syntax check, refactoring

CDS Tools: Create/manage Core Data Services views

SQL Console: Run/test SQL queries on SAP HANA

Debugger: Set breakpoints, inspect variables, analyze flow

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Classic ABAP vs RAP

- **Classic ABAP**

UI: SAP GUI-based interfaces

Logic: Function Modules, Reports, Classical Dynpros

Data Access: Direct table access using Open SQL

Architecture: Typically tightly coupled, monolithic designs

- **RAP (RESTful Application Programming Model)**

UI: Designed for Fiori/UI5 or REST-based services

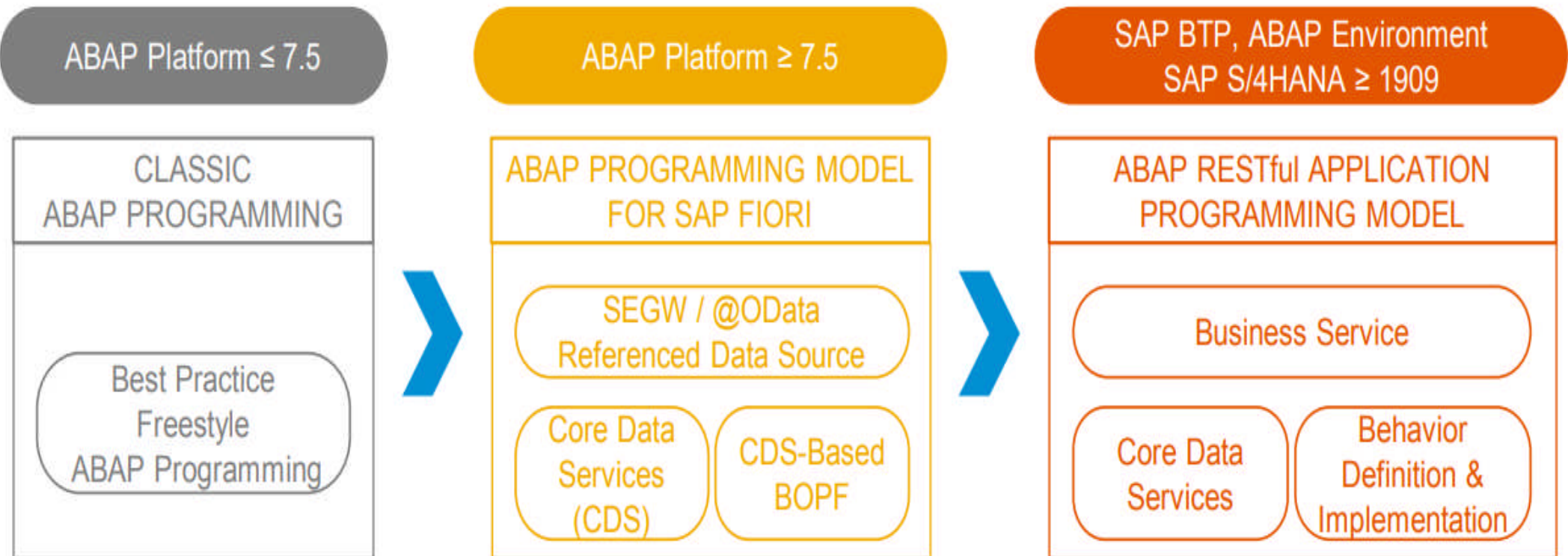
Logic: CDS Views, Behavior Definitions, Service Bindings

Data Access: Layered access using CDS with annotations

Architecture: Cloud-ready, modular, clean core approach

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Classic ABAP vs RAP



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ABAP Strict

- **What it is:**

- A restricted subset of the ABAP language

- Specifically designed for SAP BTP ABAP Environment

- Enforces clean, cloud-compliant development

- Removes access to obsolete or unsafe language elements

- **Why it matters:**

- Prevents direct access to unauthorized or sensitive repository objects

- Ensures code security, stability, and upgrade compatibility

- Aligns with cloud development best practices

- Encourages modular, service-oriented architecture

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RAP Overview

Key Layers:

- **CDS + Behavior:**

Core Data Services (CDS): Define the data model and business logic.

Behavior Definition: Describe allowed operations (e.g., create, update, delete) and business rules

- **Service Definition:**

Exposes selected CDS entities and behaviors as a service.

Acts as a contract between data model and consumers.

- **Projection:**

Defines a simplified or tailored view of the service for specific use cases or UIs.

Helps in customizing fields and behavior for end users.

- **Binding:**

Connects the service projection to a specific protocol (e.g., OData).

Makes the RAP service accessible for UI or external consumption.

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RAP Artifacts

- **CDS Views (Core Data Services):**
 - Define the data model and relationships.
 - Include annotations for UI, behavior, and service exposure.
 - Can be composed into multiple layers (interface, consumption).
- **Behavior Definition:**
 - Specifies the business behavior of CDS entities (e.g., Create, Update, Delete, Validate).
 - Includes behavior implementations in ABAP classes.
 - Enables draft handling, validations, and actions.
- **Service Definition & Binding:**
 - Service Definition:** Selects and exposes CDS views and behaviors as a service.
 - Service Binding:** Publishes the service using a protocol (e.g., OData V4).
 - Makes the application consumable via UI (e.g., Fiori Elements) or APIs.

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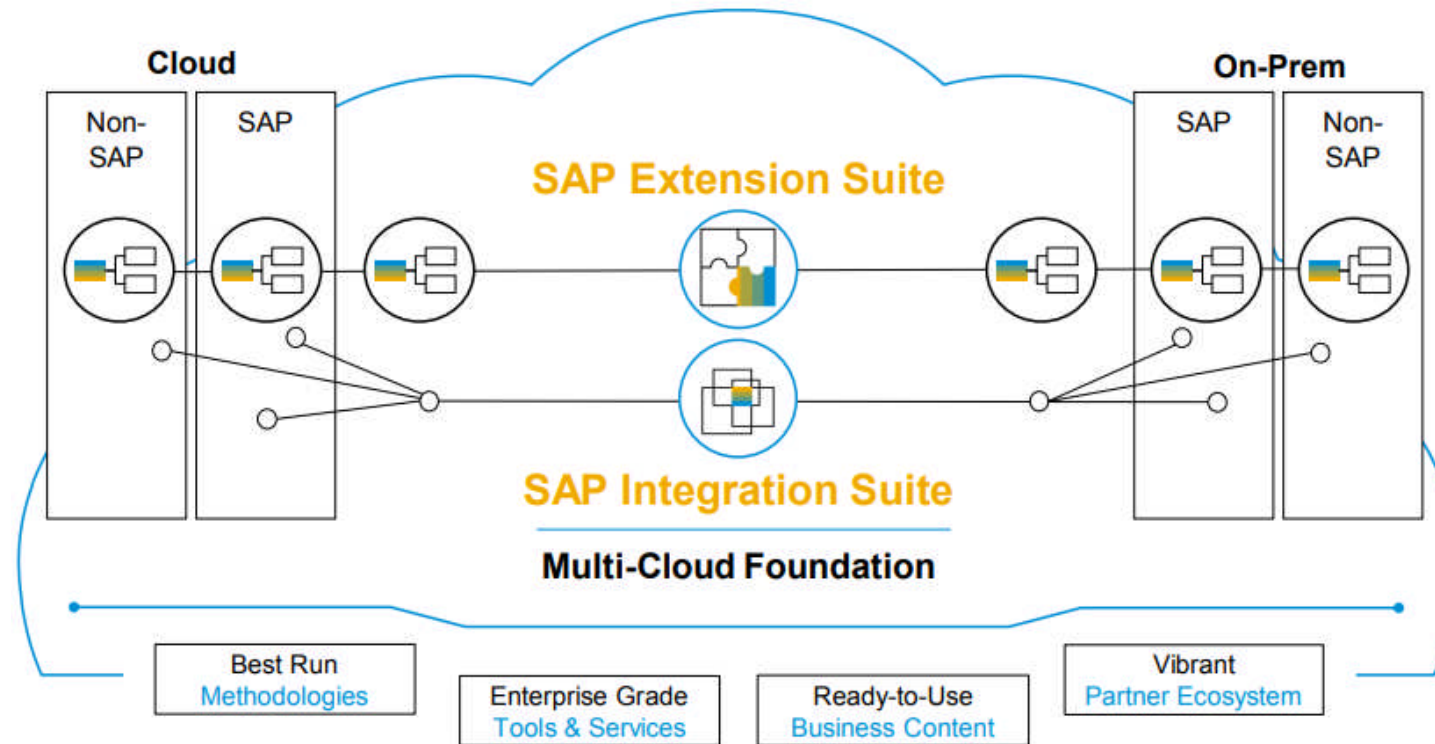
SAP Extension Suite, Integration Suite

Extension Suite

- **Capabilities:**
 - Digital Experience
 - Process Automation
 - Development Efficiency

Integration Suite

- **Features:**
 - API Business Hub
 - Cloud Integration
 - Event Mesh



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Creating A Future-ready Workforce

In-App Extensions, Side-by-Side Extensions

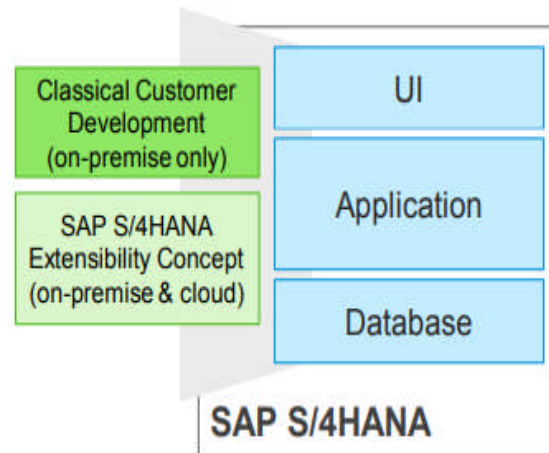
In-App Extensions

- **Details:**
 - UI adaptation
 - Custom fields
 - Business logic (BAdIs)

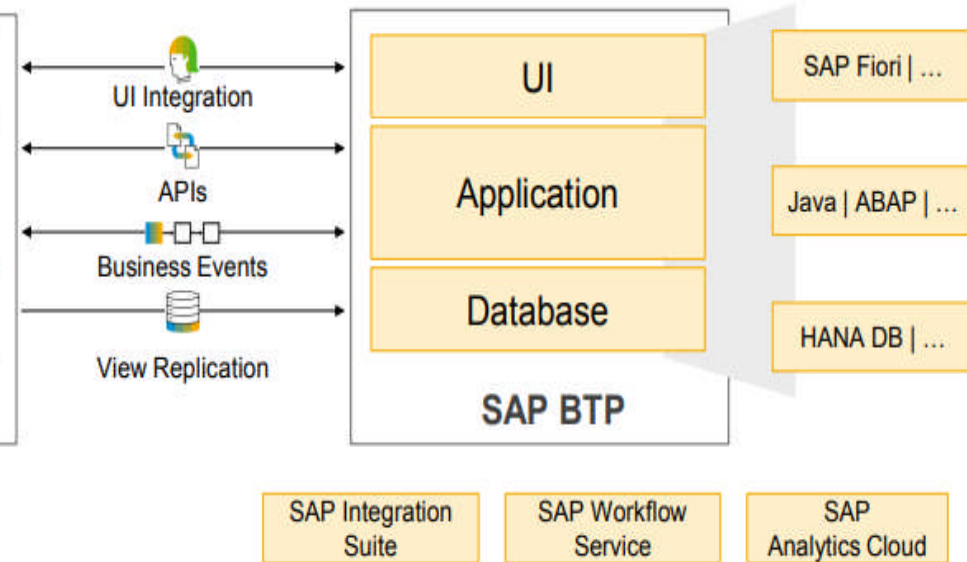
Side-by-Side Extensions

- **Details:**
 - Built on BTP
 - Uses OData, RAP, CAP

In-App Extensibility



Side-by-Side Extensibility



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Example Architecture

- **Scenario:**

Full-stack custom extension application built on **SAP BTP**, designed to extend the functionality of **SAP S/4HANA** without modifying the core system.

- **Includes:**

- Custom UI:**

- Built with SAP Fiori/UI5
 - Runs on SAP BTP Launchpad
 - Connects via OData/REST APIs

- Business Service:**

- Developed using ABAP RAP
 - Contains data model, logic, and service layer
 - Exposed via OData V4

- Data Replication:**

- Syncs data from S/4HANA to BTP
 - Uses Cloud Connector or Integration Suite
 - Enables safe extension without core changes

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Creating A Future-ready Workforce



Hands-on Demo Summary

- **Steps:**
 - Open Eclipse
 - Connect to Cloud
 - Use RAP + OData + CDS

Key SAP BTP Services

- **Tools:**
 - BAS, Workflow, API Hub, IRPA, SAP AI Core

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IaaS vs PaaS

- **IaaS (Infrastructure as a Service):**

Provides virtualized hardware resources like compute power, storage, and networking.

Managed by cloud providers like AWS, Azure, GCP.

You manage the OS and applications; the provider manages infrastructure.

Example: Running a virtual machine or database on AWS.

- **PaaS (Platform as a Service):**

Provides a complete development and runtime environment in the cloud.

You focus on building apps; the platform (e.g., SAP BTP) handles infrastructure, OS, runtime, and scalability.

Ideal for developing, deploying, and managing applications without dealing with low-level resources.

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<https://www.freepik.com/>



ABAPGit Overview

- **Purpose:**
 - Provides **Git-based version control** for ABAP development.
 - Enables **source code management**, collaboration, and change tracking in ABAP systems.
- **Key Highlights:**
 - Open-source tool, works in on-premise and SAP BTP ABAP Environment.
 - Allows developers to **push/pull ABAP objects** (classes, programs, function modules, etc.) to/from Git repositories.
 - Facilitates team collaboration, transport-independent deployments, and continuous integration.
 - Supports integration with platforms like GitHub, GitLab, Bitbucket.
- **Why It Matters:**
 - Brings modern DevOps practices to ABAP development.
 - Helps manage code history, versioning, and distributed development efficiently.

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Benefits of RAP

- **Cloud-optimized:**
Built for SAP BTP; follows clean-core and modern cloud principles.
- **Tooling:**
Uses Eclipse with ABAP Development Tools (ADT); no SAP GUI needed.
- **Rapid ramp-up:**
Easier learning curve with guided tools, templates, and consistent design patterns.

Challenges of Extensions

- **Risks:**
Upgrade Issues: Custom code may break during updates
Complexity: Hard to manage and maintain extensions
- **Solution:**
SAP Extensibility Framework ensures upgrade-safe, clean, and manageable extensions on SAP BTP.

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Summary

- **SAP BTP (Business Technology Platform)** provides a unified environment for data management, application development, analytics, and intelligent technologies.
- Developers work within a structured account model involving **Global Accounts, Subaccounts, and Spaces** to manage cloud resources.
- The **ABAP environment in BTP** enables running and developing ABAP programs natively in the cloud using the **Cloud Foundry** runtime.
- A stricter version of the ABAP language, called **ABAP “strict”**, is used in cloud environments to enforce security and clean development practices.
- All ABAP development in the cloud is done using **Eclipse with ABAP Development Tools (ADT)**, not the traditional SAP GUI.



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Continued...

- The **ABAP RESTful Application Programming Model (RAP)** is the recommended programming model for building cloud-ready SAP Fiori applications and APIs.
- RAP introduces modern development artifacts such as **CDS Views, Behavior Definitions, and Service Bindings** to streamline business logic implementation.
- The **SAP Extension Suite** helps developers create new applications and extensions efficiently using categorized services for UI, process automation, and development.
- **Extensibility in SAP S/4HANA** is enabled via **In-App** (key-user) and **Side-by-Side** (cloud-based) approaches, with the latter leveraging BTP and RAP.



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Quiz

1. What are the four core capabilities of SAP Business Technology Platform (BTP)?

- a) Business Processes, APIs, UI Design, Debugging
- b) Database, Integration, Analytics, Intelligent Technologies
- c) UI Design, API Management, Git Integration, Testing
- d) Security, Extensibility, Automation, Maintenance



Answer: B

Database, Integration, Analytics, Intelligent Technologies



Quiz

2. Which tool is used for developing ABAP applications in SAP BTP?

- a) SAP GUI
- b) Eclipse with ADT
- c) Visual Studio Code
- d) SAP Web IDE



Answer: B
Eclipse with ADT



Quiz

3. What is ABAP “strict”?

- a) A runtime monitoring tool
- b) A deprecated version of ABAP
- c) A restricted ABAP version used in the cloud
- d) An SAP GUI add-on



Answer: C

A restricted ABAP version used in the cloud



Quiz

4. What is the purpose of the Behavior Definition in RAP?

- a) To define user interface layout
- b) To manage database triggers
- c) To control the business logic and operations like create/update
- d) To manage SAP system updates



Answer: C

To control the business logic and operations like create/update



Quiz

5. Which of the following are valid extensibility options in SAP S/4HANA?

- a) Cloud-only and Hybrid
- b) Modular and Monolithic
- c) Internal and External
- d) In-App and Side-by-Side



Answer: D

In-App and Side-by-Side

Thank You